Chem(A)-II

c) Define octahedral distortion and ligand. [2]

Group C

Long Answer Questions [3 × 8 = 24]

20. Gibbs free energy is a thermodynamic function used to predict spontaneity of a process.
    a) Discuss the criteria of spontaneity, non-spontaneity and equilibrium of exothermic and endothermic reactions based on the basis of free energy change and entropy change. [3]
    b) State Hess's law of constant heat summation. Write any two applications of this law. [2]
    c) Calculate the heat of combustion of glucose from the following data. [3]
       i) $C(s) + O_2(g) \rightarrow CO_2(g)$, $\Delta H = -395$ KJ
       ii) $H_2(g) + \frac{1}{2} O_2(g) \rightarrow H_2O(l)$, $\Delta H = -269$ KJ
       iii) $6C(s) + 6H_2(g) + 3O_2(g) \rightarrow C_6H_{12}O_6(s)$, $\Delta H = -1169$ KJ

OR

    a) The standard electrode potential of silver and zinc are given below,
       $Ag^+ + 2e^- \rightarrow Ag$, $E^\circ_{Ag^+/Ag} = +0.8V$
       $Zn \rightarrow Zn^{++} + 2e^-$, $E^\circ_{Zn^{++}/Zn} = +0.76V$
       i) Write a cell notation [1]
       ii) Calculate the cell potential for the given cell at 25°C. [1]
       iii) Can you stir $AgNO_3$ solution With a Zinc rod. Provide a reason for your answer. [2]
       iv) Calculate ΔG° for a cell. [1]
       v) Why can KCl not be used as an electrolyte in a salt bridge for this specific cell? [1]
    b) Secondary cells act as both electrolytic and voltaic cells. Justify with suitable reactions. [2]
21. Differentiate between [4 x 2=8]
    a) Homo-polymer and co-polymer giving an example of each.
    b) OPC and PPC cement.
    c) Nuclear reaction and chemical reaction
    d) Fibrous and non-fibrous raw materials for the paper production.
22. Compound (A) is a primary alcohol that gives positive iodoform test and reacts with calcium- chloro hypochlorite paste to give compound (B) which is used as an anaesthetic agent.
    a) Identify compound (A) and (B) with the reactions involved. [1+1]
    b) Why compound (B) is stored in dark brown bottles? [2]
    c) What happens when compound (B) is reacted with acetone? [1]
    d) How is compound (A) distinguished from propan-2-ol by using Victor Maeyer's method? [2]
    e) Convert compound (B) into o-salicylaldehyde? [1]

OR

Vinegar is at least 4% acetic acid by volume making acetic acid the main component of vinegar a part from water.

    a) Compare the acidic strength of acetic acid with formic acid and chloroacetic acid with reason. [1+1]
    b) What happens when acetic acid is reacted with phosphorus pentoxide [1]
    c) Convert ethanoic acid to methanol. [2]
    d) When acetic acid is heated with ethanol in presence of conc. H_2SO_4 forms compound (A) which on condensation with sodium salt of acetic acid gives compound (B)
       i. Identify (A) and (B) with reaction involved. [2]
       ii. What happens when (A) is reacted with ammonia. [1]

THE END

Chem(A)-I

Kathmandu Model Secondary School

Imadol, Lalitpur

Practice Question

Subject: Chemistry [3021] | **SET A** | **Full Marks: 75**

Grade: XII, Science | **Time: 3 hrs.** | **Pass Marks: 30**

Group B

Short Answer Questions. **[8 × 5 = 40]**

12. Experimental data obtained by titrating decinormal solution of sodium carbonate against hydrochloric acid solution are given below.

Exp. No.	Volume of Na_2CO_3 (mL)	Volume of HCl burette reading (mL)	
		Initial	Final
1.	10	0.0	10.9
2.	10	0.0	10.5
3.	10	0.0	10.5

a) Which indicator is suitable for above titration? Explain with proper reason. [2]
b) What volumes of water must be added to 250 mL of HCl solution (used in above titration) to make it exactly centi-normal? [2]
c) Why is sodium carbonate considered a primary standard substance? [1]

13. The following rate data were obtained at 303K for the reaction $2A + B \rightarrow C + D$

Exp. No.	[A] mol L^{-1}	[B] mol L^{-1}	Initial rate of formation of [D] mol L^{-1} min^{-1}
1	0.1	0.1	6.0×10^{-3}
2	0.3	0.2	7.2×10^{-2}
3	0.3	0.4	2.88×10^{-1}
4	0.4	0.1	2.4×10^{-2}

a) Determine the order with respect to each reactant and overall order. [3]
b) Calculate the value of rate constant (k) along with its appropriate unit. [1]
c) Write a rate law equation. [1]

OR

a) State Ostwald's dilution law. Why is this law not applicable for strong electrolytes? [1+1]
b) Calculate the pH of a solution obtained by mixing 200 mL of pH = 12 solution with 150 mL of pH = 2 solution. [3]

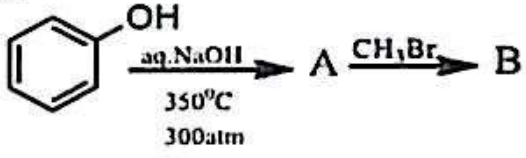
14. An organic compound 'A' obtained by the reduction of ethanenitrile turns a red litmus into blue.
a) Write the reaction involved for the formation of 'A' as described above. [1]
b) Compare the basic strength of 'A' with N-methyl aminomethane and aniline. [2]
c) Give a suitable test to distinguish 'A' from N-methyl aminomethane. [2]

OR

Nitrobenzene is a yellow oily liquid with bitter almond smell
a) Nitro-group in nitrobenzene is meta-directing in nature towards electrophilic substitution reaction. Why? [2]
b) How is nitrobenzene reduced in
i) Acidic medium
ii) Neutral medium
iii) With strong reducing agents $LiAlH_4$. [1+1+1]

15. Aliphatic aldehydes are more reactive than Benzaldehyde.
a) Give reason for the above statement. [2]
b) Show your acquaintance with Wolf Kishner Reduction. [1]
c) How would you distinguish benzaldehyde and acetaldehyde? [1]
d) What product would you expect when formaldehyde is treated with phenol in acidic medium? [1]

16. Aliphatic and aromatic ethers can be prepared by Williamson's ether synthesis.
a) Write down the preparation reaction for methoxymethane by same method? [1]
b) Explain, why Anisole is prepared by using sodium phenolate and methyl iodide but cannot be prepared using sodium methoxide and chlorobenzene. [2]
c) Identify A and B. [2]



17. a) Make a correct sequence of chemical reactions using suitable conditions and reagents of the following compounds. [3]
Benzene diazonium chloride, Aniline, Chlorobenzene, Benzylamine, Benzonitrile.
b) Phenols are ortho-para directing in nature towards electrophilic substitution reaction, why? [2]

18. Silver (commonly called Chandi) is used as ornamental purpose.
a) Define spongy silver. Discuss the cyanide process for the extraction of silver. [1+3]
b) Why is $AgNO_3$ known as lunar caustic? [1]

19. The shape of complex ion depends upon the type of hybridization involved, co-ordination number (no. of ligands) and size of the ligands attached.
a) Cu^{++} complexes are colored while those of Zn^{++} are colorless. Explain with reason. [2]
b) Crystal field splitting does not occur in absence of ligands, why? [1]